

Week 5: LLM Training & Fine-tuning

Day 1: Mini project briefing

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How this afternoon works

Pick an idea, firm it up, and submit a one-page proposal — you'll present it on Friday.

1

Pick an idea

From the gallery — or bring your own domain.

2

Firm it up

Name the gap, define the eval, scope it to one afternoon.

3

Submit a proposal

One page, six items.

This afternoon: choose a project and draft a one-page proposal. Ask us anything while you scope.

 WHAT IT IS

Customize a model — and defend the decision

A team project that puts the whole week together on a problem you choose.

THE SHAPE OF IT

- Pick a task where a base model falls short.
- Add the missing capability — in the weights, not RAG.
- Measure honestly: what improved, and what it cost.
- Recommend deploy / iterate / hold — and defend it.

WHAT MATTERS

- The reasoning, not the highest score.
- A one-page decision brief.
- A 5-minute presentation.
- Your working notebook + the numbers behind it.

A defensible “we recommend hold” beats a great number you can’t explain.

Decide it together, own it together

One project per team. First agree your topic — then split the work so everyone owns a piece.

1 · AGREE YOUR TOPIC

- Pick one idea as a team — from the galleries or your own.
- Run it through the four checks together.
- One project, one proposal, one shared goal.

2 · SPLIT THE WORK

- Data & eval — build the test set, own the metric.
- Modelling — run the customization, track what you try.
- Analysis & write-up — error analysis + the decision brief.
- Coordinator — keep scope tight, lead the lightning talk.

Roles can overlap on a small team — the point is that everyone contributes and no one carries it alone.

Teach it something new

Pick a task where a base model falls short on knowledge or behaviour. A few starting points:

CLASSIFY & ROUTE

- Support-ticket triage (category + priority)
- Incident / field-report triage
- Content-moderation labelling
- Bug reports → severity + component
- Email / intent routing

EXTRACT TO STRUCTURE

- Receipts / invoices → JSON
- Meeting notes → action items
- Job posts → skills schema
- Case notes → structured fields
- Contracts → key clauses

INJECT KNOWLEDGE

- Company / product facts (closed-book)
- Internal policy or HR handbook
- API / library documentation
- Game or fictional-world lore
- Course syllabus & FAQ

These all need new knowledge or behaviour in the weights — the kind of gap fine-tuning is built for.

Change how it answers

Same facts, different behaviour — shape the model's style, preferences, or reasoning.

MATCH A STYLE / VOICE

- Brand-voice rewriter
- Formal ↔ friendly tone
- Plain-language simplifier
- Domain house-style answers

SET A PREFERENCE

- Concise over rambling
- Grounded / cited answers
- Graceful, safe refusals
- One consistent output format

TEACH A METHOD

- Domain step-by-step reasoning
- A standard explanation template
- Checklist-driven answers
- Show-your-working style

Best of all — bring your own domain. These are starting points, not a fixed menu. Anything you care about works, if it passes the checks.

Turn an idea into a project — four checks

Run any idea — from the gallery or your own — through these four. If it clears them, it's a good project.

1

Start from a real gap

Where does a base model fall short on your task?
No gap, no project.

2

Match technique to the gap

Knowledge or behaviour → SFT (LoRA).
Style or preference between valid outputs → DPO.
You'll learn each this week; just name the fit for now.

3

Make it measurable

Decide your eval first.
You need a primary metric and ~20–30 labelled examples you can actually get this afternoon.

4

Scope it this afternoon

One narrow task, one metric, finishable on Friday.
Over-scoping is the #1 reason teams don't finish.

Scope it to one afternoon

Whatever the idea, the difference between finishing and not is scope.

✓ ONE-AFTERNOON SCOPE

- One narrow task, one primary metric.
- A gap a base model actually has. (open weight model)
- ~20–30 labelled examples you can get today.
- Finishable in Friday's build block.

✗ OVER-SCOPED

- Several tasks or metrics at once.
- A big new dataset to collect and label.
- A live deployment or full UI.
- “Beat GPT-4 on everything.”

When in doubt, cut. A tight scope with an honest metric beats an ambitious plan you can't finish.

Support-ticket triage

BEHAVIOUR

Read a customer ticket and output the right category + priority for routing.

THE GAP

The base model guesses the category and mis-calls priority — inconsistent, and it misses the urgent ones.

THE FIT

Fine-tune the behaviour (SFT / LoRA) so it applies your routing rules consistently.

HOW YOU'D MEASURE IT

Both fields correct together, on ~25 labelled tickets spanning easy, boundary and rare cases, with urgent recall tracked on its own.

SCOPE

In: One ticket → category + priority, on a small labelled set.

Out: Free-text replies, new categories, live routing.

Why it's good: a measurable metric, a technique matched to the gap, and a scope you can finish.

Closed-book product assistant

KNOWLEDGE

Answer questions about your product's facts and policies from memory — no retrieval.

THE GAP

The base model has never seen your product, so it either can't answer or invents facts.

THE FIT

Inject the facts into the weights (SFT / LoRA) so they're recalled closed-book.

HOW YOU'D MEASURE IT

Three sub-metrics: recall of taught facts, abstaining on unknowns (no hallucination), and no drop on general knowledge.

SCOPE

In: ~20 facts, answered closed-book and measured three ways.

Out: A retrieval system, web lookup, or expanding the fact set.

Why it's good: a measurable metric, a technique matched to the gap, and a scope you can finish.

Brand-voice rewriter

PREFERENCE

Rewrite plain answers into your house voice — same meaning, right tone.

THE GAP

The content is fine but the tone is generic; two valid answers differ only in style.

THE FIT

Teach the preference between valid outputs (DPO) from chosen-vs-rejected pairs.

HOW YOU'D MEASURE IT

Win-rate of the tuned voice vs the base model on held-out prompts, judged against a short style rubric.

SCOPE

In: One domain, one house style, a small set of preference pairs.

Out: Multiple brands, multi-turn chat, a live writing tool.

Why it's good: a measurable metric, a technique matched to the gap, and a scope you can finish.

One page, six questions

A scoping check, not a deliverable: the point is to define your eval before you build.

THE SIX QUESTIONS

- Domain & task — what comes in, what the model outputs.
- Why weights, not RAG — one line.
- Expected gap — knowledge, behaviour, or preference?
- Eval plan — primary metric + how you'll get ~20–30 labelled examples.
- Data plan — provided / synthetic / hand-written.
- Scope check — what you'll attempt vs explicitly not do.

IT PASSES IF...

- The eval is defined, with a primary metric.
- The technique matches the expected gap.
- The data plan is realistic for one afternoon.
- The scope is honestly one-afternoon-sized.

From today to Friday

The week is one continuous build — each lab feeds your project.

Mon PM

Briefing, choose a project, **draft proposal submitted to xsite dropbox by today**

Tue–Thu

Labs run your chosen technique; test/refine your plan

Fri PM

Build the project (~2.5 h) + presentation (5 min each)
Submit to xsite dropbox: Presentation slides, runnable code with user manual

Final presentation

You should answer

- **The problem** — what task did you pick, and what does the model take in / put out?
- **The gap** — where did the base model fall short? Show one concrete failure
- **What you did** — which technique (SFT / DPO) and why it fits the gap; keep it high-level
- **The results** — your primary metric, before vs after; be honest about what didn't improve
- **Error analysis** — one or two cases it still gets wrong, and why
- **Your recommendation** — deploy / iterate / hold, and defend the call

Presentation

- ✓ **Format:** 5-minute per team, followed by 2 minutes of Q&A
- ✓ **Slides:** 5–6 slides maximum; every team member speaks
- ✓ **Time it:** practice once — going over time counts against you
- ✓ **Submission to xsite:** slides, notebook, data in a zip file by this Friday night (11:30PM)

The mini-project is 60% of your grade

Three components — weighted toward defensible work, not the proposal paperwork.

10%

Project proposal

One page, six prompts. Eval defined, technique matched to the gap, scope honest. Due Today.

10%

Presentation

A 5-minute presentation in Friday afternoon: the problem, what you did, honest results, and your recommendation.

40%

Quality — code & results

Working notebook, sound method, honest measurement. Well prepared data. Due Friday night.

= 60% of the final grade

Let's work on proposal now

CHECKLIST

- Pick the idea.
- Discuss with team
- Run it through the four checks.
- Open the proposal template; draft the six prompts.

A STRONG PROPOSAL...

- ✓ Names a clear, measurable primary metric.
- ✓ Matches the technique to the gap.
- ✓ Is scoped to finish in one afternoon.